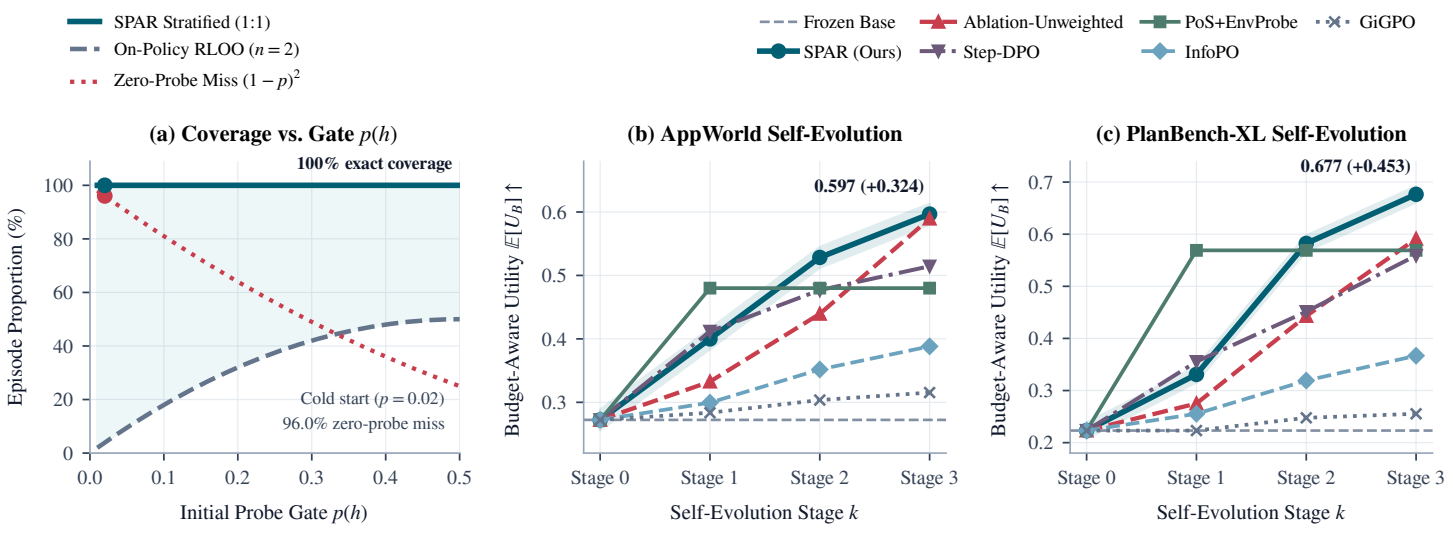
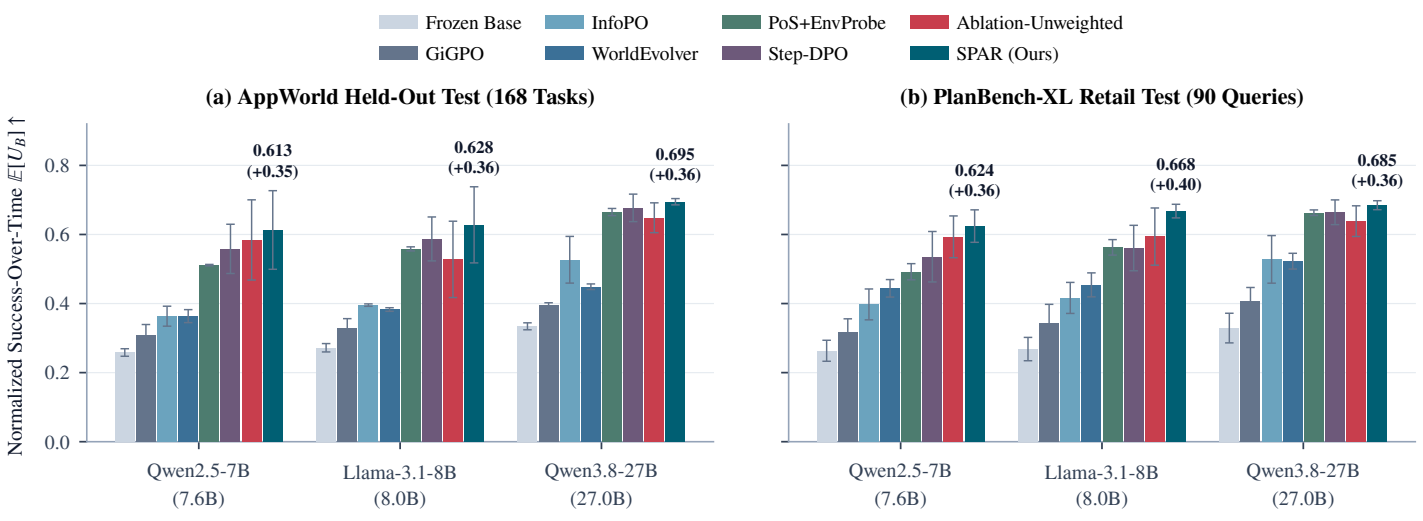
Method	AppWorld `test_normal`, 168 tasks				PlanBench-XL `test`, 90 queries			
	$\mathbb{E}[U_B]\uparrow$	$\Delta_{\text{evo}}\uparrow$	$\mathbb{E}[Y](\%)\uparrow$	$\Delta_{\text{use}}\uparrow$	$\mathbb{E}[U_B]\uparrow$	$\Delta_{\text{evo}}\uparrow$	$\mathbb{E}[Y](\%)\uparrow$	$\Delta_{\text{use}}\uparrow$
Qwen2.5-7B-Instruct (7.6B, $d = 3584$)								
Frozen Base	0.2583 ± 0.0109	$+0.0000$	34.92	$+0.2042$	0.2633 ± 0.0304	$+0.0000$	35.93	$+0.1167$
AREW / T3	0.2771 ± 0.0127	$+0.0188$	37.70	$+0.2333$	0.2809 ± 0.0218	$+0.0176$	38.52	$+0.1167$
GiGPO	0.3094 ± 0.0299	$+0.0512$	42.46	$+0.2625$	0.3163 ± 0.0395	$+0.0530$	43.70	$+0.1361$
InfoPO	0.3634 ± 0.0289	$+0.1052$	50.40	$+0.3208$	0.3975 ± 0.0447	$+0.1341$	55.56	$+0.2194$
WorldEvolver	0.3636 ± 0.0188	$+0.1053$	50.40	$+0.3694$	0.4441 ± 0.0253	$+0.1807$	62.59	$+0.3208$
PoS+EnvProbe	0.5112 ± 0.0021	$+0.2529$	72.42	$+0.4083$	0.4921 ± 0.0234	$+0.2288$	70.00	$+0.3951$
Step-DPO	0.5583 ± 0.0713	$+0.3000$	79.37	$+0.4958$	0.5356 ± 0.0730	$+0.2723$	75.93	$\mathbf{+0.4142}$
Ablation-Unweighted	0.5842 ± 0.1161	$+0.3260$	**88.49**	$+0.3912$	0.5932 ± 0.0605	$+0.3299$	90.37	$+0.2403$
SPAR (Ours)	$\mathbf{0.6130} \pm 0.1138$	$\mathbf{+0.3547}$	**88.49**	$\mathbf{+0.5033}$	$\mathbf{0.6243} \pm 0.0470$	$\mathbf{+0.3610}$	**91.48**	$+0.3724$
Llama-3.1-8B-Instruct (8.0B, $d = 4096$)								
Frozen Base	0.2610 ± 0.0099	$+0.0000$	35.32	$+0.2139$	0.2683 ± 0.0226	$+0.0000$	36.67	$+0.1167$
AREW / T3	0.2744 ± 0.0141	$+0.0134$	37.30	$+0.2333$	0.2933 ± 0.0326	$+0.0249$	40.37	$+0.1167$
GiGPO	0.3148 ± 0.0338	$+0.0538$	43.25	$+0.2819$	0.3189 ± 0.0350	$+0.0506$	44.07	$+0.1458$
WorldEvolver	0.3688 ± 0.0123	$+0.1078$	51.19	$+0.3694$	0.4441 ± 0.0253	$+0.1757$	62.59	$+0.3208$
InfoPO	0.3838 ± 0.0232	$+0.1227$	53.37	$+0.3500$	0.4127 ± 0.0517	$+0.1444$	57.78	$+0.2486$
PoS+EnvProbe	0.5112 ± 0.0021	$+0.2501$	72.42	$+0.4083$	0.5067 ± 0.0044	$+0.2384$	72.22	$+0.4142$
Step-DPO	0.5782 ± 0.0725	$+0.3171$	82.34	$+0.5153$	0.5567 ± 0.0680	$+0.2883$	79.26	$+0.3746$
Ablation-Unweighted	0.6258 ± 0.0663	$+0.3647$	96.23	$+0.4536$	0.6197 ± 0.0442	$+0.3514$	95.19	$+0.2806$
SPAR (Ours)	$\mathbf{0.6592} \pm 0.0495$	$\mathbf{+0.3982}$	**96.63**	$\mathbf{+0.5453}$	$\mathbf{0.6507} \pm 0.0386$	$\mathbf{+0.3824}$	**95.93**	$\mathbf{+0.3962}$

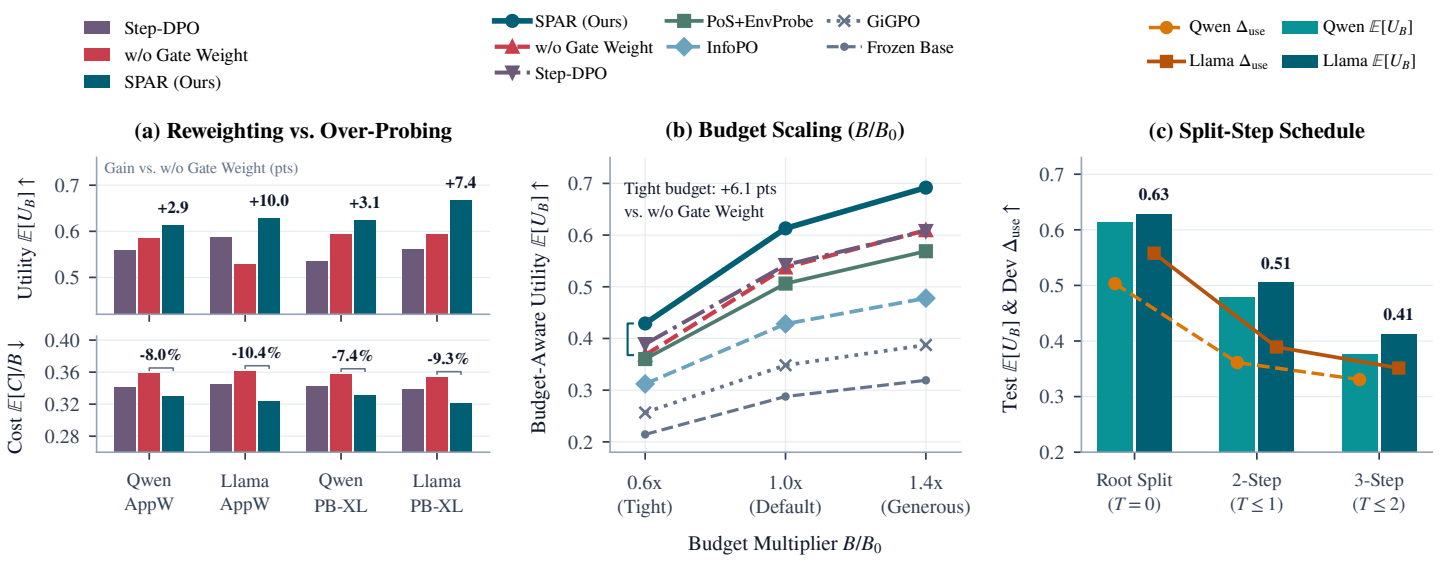
Table 1: Main performance comparison on held-out test sets across two open-weight LLM backbones (3 seeds, Mean $\pm$ Std). $\mathbb{E}[U_B]$ denotes normalized success-over-time utility, Δ_{evo} denotes self-evolution gain over Frozen Base, $\mathbb{E}[Y]$ is native terminal verification success rate, and Δ_{use} measures development-set masked-feedback utilization.

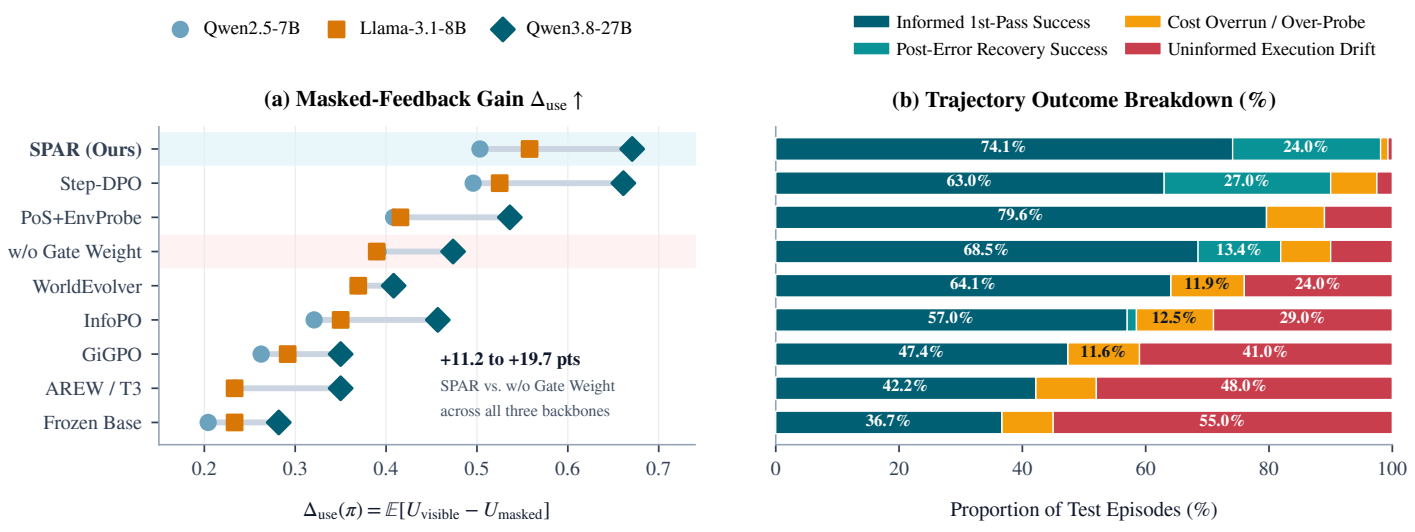
Diagnostic metric ($p = 0.4$)	Two-sample RLOO	SPAR
Expected objective $J(\theta_0)$	$58/125$ $= 0.4640$	$58/125$ $= 0.4640$
Analytical $\nabla_\theta J(\theta_0)$ bias	0.0000	$\mathbf{0.0000}$
Root-gate variance diff.	Baseline	$\mathbf{-0.0164736}$
Total gradient variance $\sum_i \mathrm{Var}(\widehat{g}_i)$	0.0711904	$\mathbf{0.0395168}$ (-44.5%)
Expected pair cost $\mathbb{E}[C]/B$	0.4800	0.5000
Variance $\times$ cost product	$0.0341714\,B$	$\mathbf{0.0197584\,B}$ (-42.2%)
Cold-start miss rate ($p = 0.02$)	96.04%	$\mathbf{0.00}\%$

Table 2: Exact rational enumeration verification of Theorem 3.1 and Theorem 3.2 on the canonical two-stage partially observable benchmark.


SPAR Stratified (1:1)
On-Policy RLOO ($n = 2$)
Zero-Probe Miss $(1-p)^2$
Frozen Base
Ablation-Unweighted
PoS+EnvProbe
GiGPO
SPAR (Ours)
Step-DPO
InfoPO
(a) Coverage vs. Gate $p(h)$
100% exact coverage
Cold start ($p = 0.02$)
96.0% zero-probe miss
Episode Proportion (%)
Initial Probe Gate $p(h)$
(b) AppWorld Self-Evolution
0.597 (+0.324)
Budget-Aware Utility $\mathbb{E}[U_B] \uparrow$
Stage 0
Stage 1
Stage 2
Stage 3
Self-Evolution Stage k
(c) PlanBench-XL Self-Evolution
0.677 (+0.453)
Budget-Aware Utility $\mathbb{E}[U_B] \uparrow$
Stage 0
Stage 1
Stage 2
Stage 3
Self-Evolution Stage k


Frozen Base
InfoPO
PoS+EnvProbe
Ablation-Unweighted
GiGPO
WorldEvolver
Step-DPO
SPAR (Ours)
(a) AppWorld Held-Out Test (168 Tasks)
(b) PlanBench-XL Retail Test (90 Queries)
Normalized Success-Over-Time $\mathbb{E}[U_B] \uparrow$
0.0
0.2
0.4
0.6
0.8
0.613 (+0.35)
0.628 (+0.36)
0.695 (+0.36)
0.624 (+0.36)
0.668 (+0.40)
0.685 (+0.36)
Qwen2.5-7B (7.6B)
Llama-3.1-8B (8.0B)
Qwen3.8-27B (27.0B)


Step-DPO
w/o Gate Weight
SPAR (Ours)
SPAR (Ours)
w/o Gate Weight
Step-DPO
PoS+EnvProbe
InfoPO
GiGPO
Frozen Base
Qwen Δ_{use}
Llama Δ_{use}
Qwen $\mathbb{E}[U_B]$
Llama $\mathbb{E}[U_B]$
(a) Reweighting vs. Over-Probing
Gain vs. w/o Gate Weight (pts)
Utility $\mathbb{E}[U_B]\uparrow$
+2.9
+10.0
+3.1
+7.4
Cost $\mathbb{E}[C]/B\downarrow$
-8.0%
-10.4%
-7.4%
-9.3%
Qwen AppW
Llama AppW
Qwen PB-XL
Llama PB-XL
(b) Budget Scaling (B/B_0)
Tight budget: +6.1 pts
vs. w/o Gate Weight
Budget-Aware Utility $\mathbb{E}[U_B]\uparrow$
0.6x (Tight)
1.0x (Default)
1.4x (Generous)
Budget Multiplier B/B_0
(c) Split-Step Schedule
Test $\mathbb{E}[U_B]$ & Dev $\Delta_{\text{use}}\uparrow$
0.63
0.51
0.41
Root Split ($T=0$)
2-Step ($T\leq 1$)
3-Step ($T\leq 2$)


Qwen2.5-7B
Llama-3.1-8B
Qwen3.8-27B
(a) Masked-Feedback Gain Δ_{use} ↑
SPAR (Ours)
Step-DPO
PoS+EnvProbe
w/o Gate Weight
WorldEvolver
InfoPO
GiGPO
AREW / T3
Frozen Base
0.2
0.3
0.4
0.5
0.6
0.7
+11.2 to +19.7 pts
SPAR vs. w/o Gate Weight
across all three backbones
$\Delta_{use}(\pi) = \mathbb{E}[U_{visible} - U_{masked}]$
Informed 1st-Pass Success
Post-Error Recovery Success
Cost Overrun / Over-Probe
Uninformed Execution Drift
(b) Trajectory Outcome Breakdown (%)
74.1%
24.0%
63.0%
27.0%
79.6%
68.5%
13.4%
64.1%
11.9%
24.0%
57.0%
12.5%
29.0%
47.4%
11.6%
41.0%
42.2%
48.0%
36.7%
55.0%
0
20
40
60
80
100
Proportion of Test Episodes (%)